LINEAR ALGEBRA · LESSON 1

The Geometry of Linear Equations

One question, three ways to see it: the **row picture**, the **column picture** and the **matrix form** $Ax = b$.

Explained from zero on a real money question: splitting ₹10 lakh between two funds.

2 × 2 and 3 × 3 | Portfolio mix | MIT 18.06, Lecture 1

THE QUESTION

You have ₹10 lakh and two funds

Fund	Return per year	On ₹1 lakh
Nifty 50 index fund	12%	₹12 thousand
Government bond fund	7%	₹7 thousand

You want to invest **all** ₹10 lakh and earn **₹1,05,000 a year.**

THE QUESTION

How much goes into each fund?

Returns are illustrative round numbers, not forecasts.

STEP 1

Turn it into equations

Let x = ₹ lakh in Nifty and y = ₹ lakh in bonds. Each requirement is one equation:

$$\begin{aligned} x + y &= 10 && \text{budget (₹ lakh)} \\ 12x + 7y &= 105 && \text{return (₹ thousand)} \end{aligned}$$

Two equations that must both hold are a **system.** Each unknown is only multiplied by a number and added, so the equations are **linear.**

THE FUNDAMENTAL PROBLEM OF LINEAR ALGEBRA

Solve n linear equations in n unknowns.

THE BIG IDEA

Three ways to read the same equations

View	You look at	The question becomes
Row picture	one equation at a time	Where do the lines **meet**?
Column picture ★	one fund at a time	Which **mix** of fund arrows hits the target?
Matrix form	everything at once	Find x with $Ax = b$.

The ★ marks the view the whole course is built on.

STEP 2

Each fund is a vector

₹1 lakh of a fund uses ₹1 lakh of budget and earns its return. Stack the two numbers in a column:

$$\text{Nifty} = \begin{bmatrix} 1 \\ 12 \end{bmatrix} \qquad \text{Bond} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

- **Scale:** $7\,(1, 12) = (7, 84)$: ₹7 lakh of Nifty.
- **Add:** $(1, 12) + (1, 7) = (2, 19)$: ₹1 lakh in each.

LINEAR COMBINATION

Scale vectors and add them: $x\,c + y\,d$. Here, a linear combination of the funds is simply **a portfolio**.

STEP 3

Matrix form: $Ax = b$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 10 \\ 105 \end{bmatrix}}_{b}$$

Rows of A = the rules (budget, return). **Columns** of A = the funds.

By columns: $A \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}$

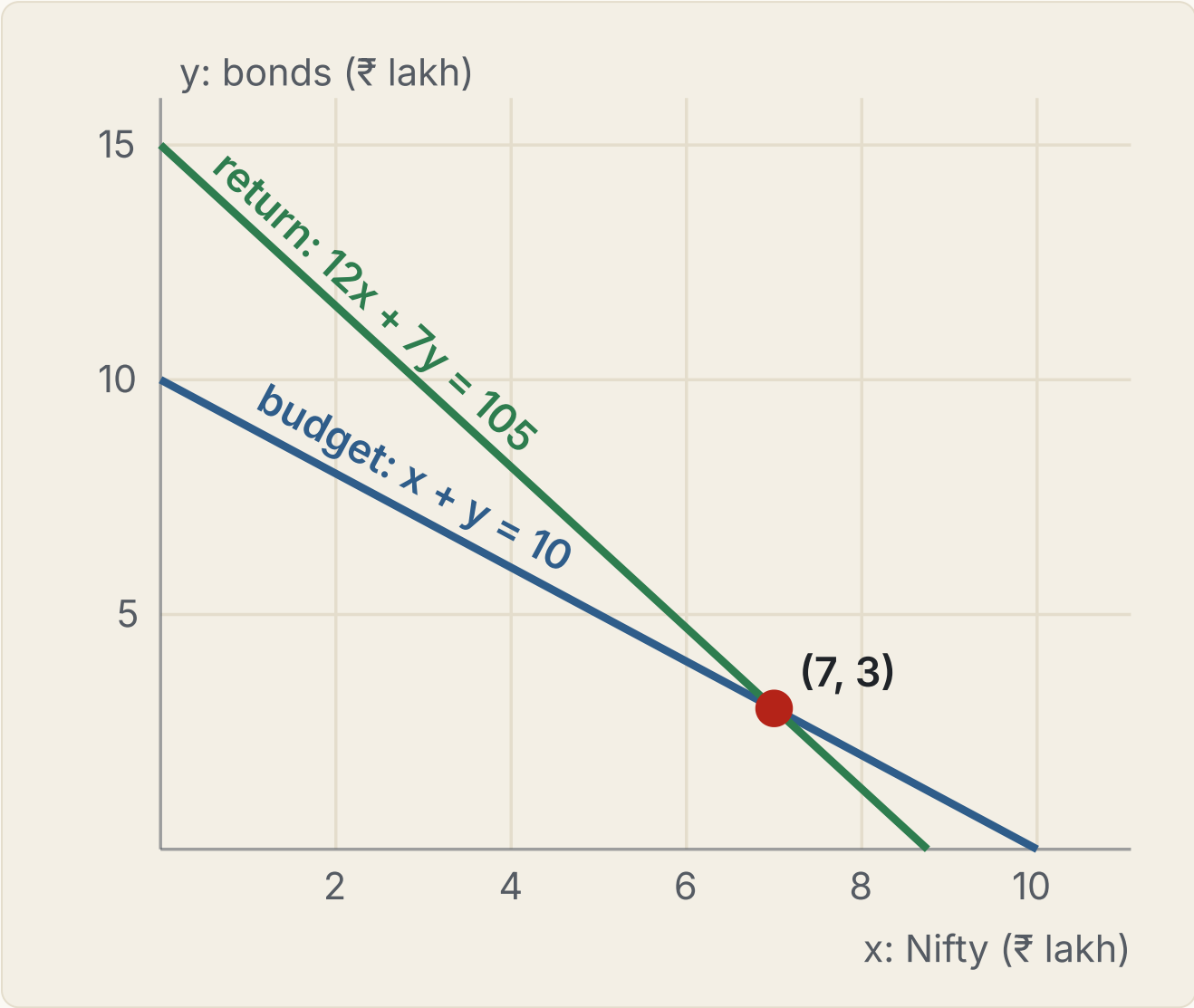
By rows: $1 \cdot 7 + 1 \cdot 3 = 10, \quad 12 \cdot 7 + 7 \cdot 3 = 105$

KEY FACT

Ax is a combination of the columns of A.

VIEW 1

Row picture: lines that cross



Each rule is a line. The answer is where they cross:

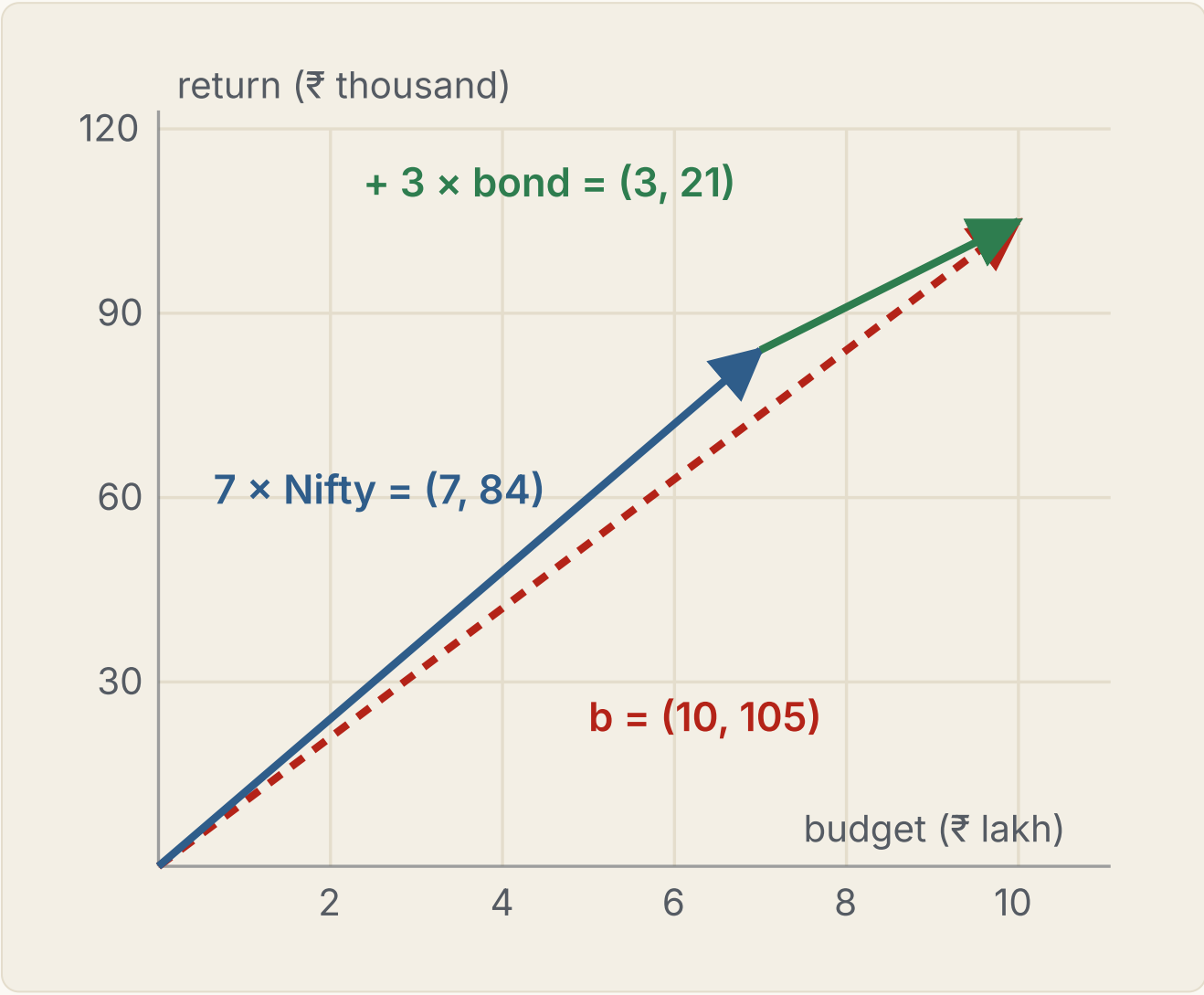
₹7 lakh Nifty, ₹3 lakh bonds.

Check: $7 + 3 = 10$ ✓ and $84 + 21 = 105$ ✓

VIEW 2 ★

Column picture: mix the arrows

$$x \begin{bmatrix} 1 \\ 12 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}$$



Same answer, seen as a **mix of fund arrows** landing exactly on the target b.

THE BIGGER QUESTION

Which targets can we reach?

The **span** of some vectors = every point you can reach by mixing them.

Funds you may use	Span	Meaning
Nifty only	a line	always exactly 12%
Bonds only	another line	always exactly 7%
Nifty + bonds	the whole plane	any budget, any return

COLUMN SPACE $C(A)$

The span of the columns of A: every target your funds can produce.

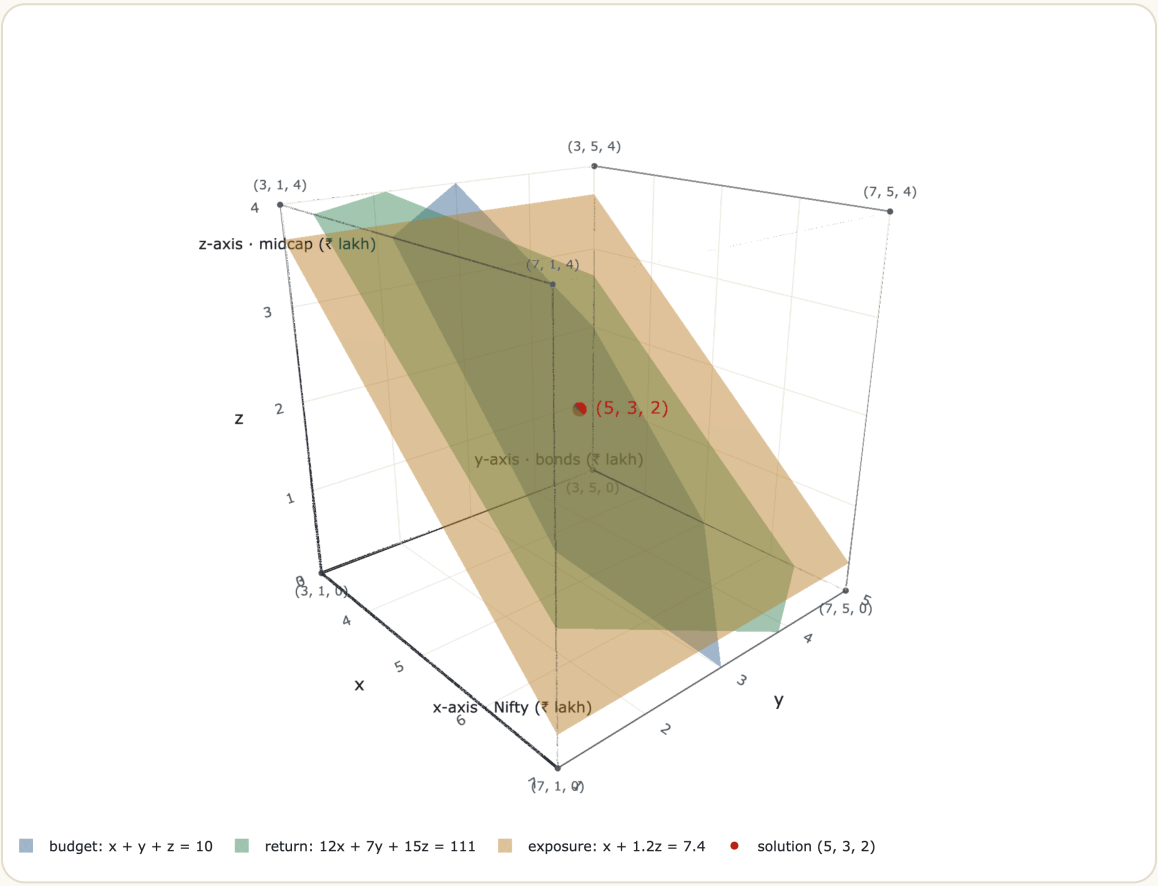
WHY IT MATTERS

$Ax = b$ has a solution **exactly when b is in $C(A)$.**

THREE FUNDS: 3 × 3

Add a midcap fund and a third rule

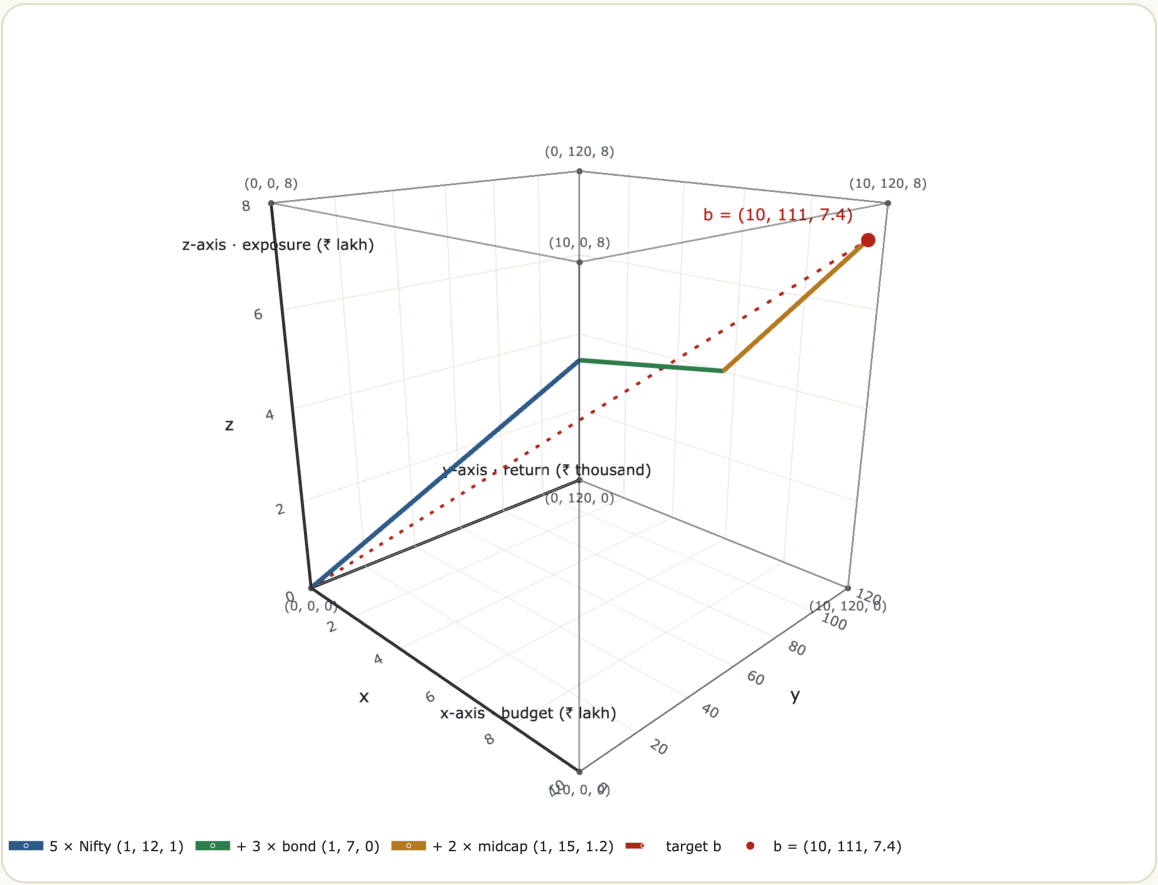
$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix} \quad \begin{matrix} \text{budget} \\ \text{return} \\ \text{market exposure} \end{matrix}$$



Row picture: three planes meet at one point: **₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap.**

Column picture in 3D

$$5\begin{bmatrix}1\\12\\1\end{bmatrix}+3\begin{bmatrix}1\\7\\0\end{bmatrix}+2\begin{bmatrix}1\\15\\1.2\end{bmatrix}=\begin{bmatrix}10\\111\\7.4\end{bmatrix}$$

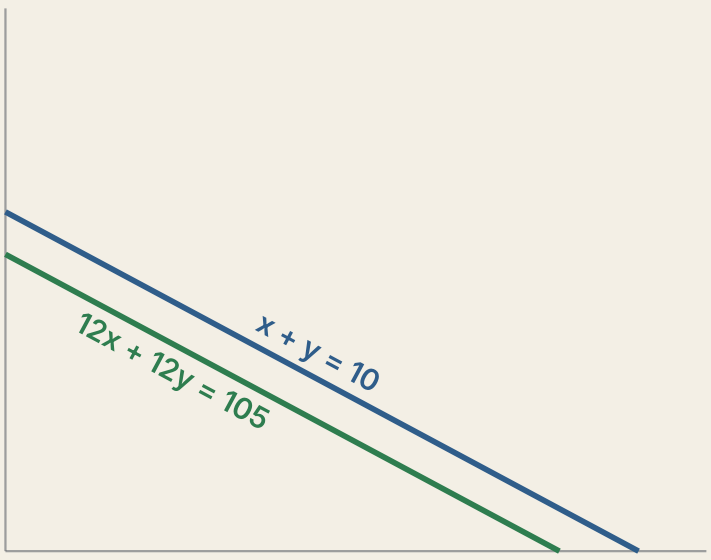


Change the target and the row picture gets three **new** planes. The column picture keeps the **same** three arrows, just mixed differently.

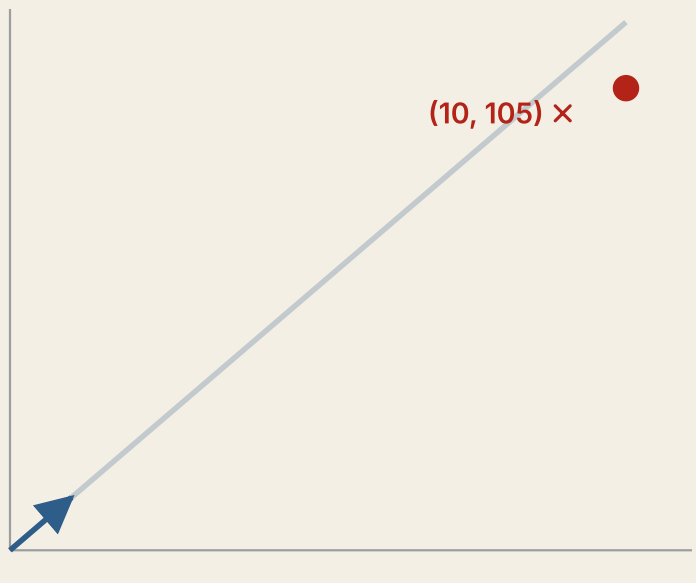
WHEN IT GOES WRONG

Singular matrices

Two Nifty funds (both 12%) point the same way:



Parallel lines: no solution.



Every mix earns 12%: the target is off the line.

SINGULAR VS INVERTIBLE

Invertible: the columns fill the space: one solution for every b. **Singular:** a fund adds no new direction (a twin fund, or a 50/50 balanced fund): no solution or infinitely many.

RECAP

Key takeaways

1. A linear system is $Ax = b$: rows are rules, columns are funds.
2. **Row picture:** lines (or planes) meet at the solution.
3. **Column picture ★:** b is a mix of the columns; the solution gives the amounts.
4. Ax is a combination of the columns of A.
5. Every target is reachable exactly when the columns fill the space (A invertible).
6. A fund built from others adds no direction: A is singular.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

Fund mixer, 3D plots you can rotate, worked examples and practice:

learn.krisarj.com

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